

**Exploring Risk and Market Sentiment in the
A-Share Communications Equipment Industry
under the Digital Economy:
An Analysis Based on A Risk Scoring and Time
Series Analysis**

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Abstract

In the last couple of years, international markets have been very volatile, influenced by many different factors, such as great economic events, policy adjustments, geopolitical tensions, and even global crises like the pandemic. This volatility has been especially pronounced in sectors like the **communications equipment industry**, which is highly affected by technological changes, such as digital transformation, 5G, and the Internet of Things. Therefore, this research will develop a relationship between market sentiment and risk analysis in the communications equipment sector of Chinese A-shares using three methodologies: **risk scoring**, **Granger causality testing**, and **ARIMA time series analysis**.

The first step in this study is data collection on the risk measurement for the communications equipment sector in the Chinese A-share market, which includes the following five types of data: turnover, volatility, beta coefficient, Sharpe ratio, and value at risk. The data standardization eliminates discrepancies in units and scales in the analysis process, which guarantees the accuracy of the following risk scoring and modeling results.

First is the computation of the risk grades of each stock using the **principal component analysis**, which simplifies the traditional risk assessment models to give the main factors of risks that will affect the respective risk scores. Starting with the five collected risk indicators, PCA extracts the meaningful risk factors behind each stock and calculates a weighted score of risks.

After calculating the risk scores, Granger causality testing is used to determine whether the variance in the risk scores has an impact on the volatility of the stock price. By analyzing the variance in the risk scores, the study quantifies the causal relationship between market sentiment fluctuations and stock price movements, with particular attention to whether changes in risk scores can act as precursors to market volatility.

The next step is an ARIMA model that tries to capture the time-series features of the fluctuations of **market sentiment** and predicts its future trends for variance of risk scores, thus reflecting variations in market sentiment within this industry.

By incorporating these, it's bound to yield an analysis of improved predictability of fluctuations in investor market sentiment.

Keywords: Market Sentiment, Risk Analysis, Communication Equipment Industry, Principal Component Analysis (PCA), Granger Causality Test, ARIMA Time Series Analysis, Investment Decision, Risk Scoring, Time Series Forecasting, Stock Market Volatility, Digital Economy

Table of Contents

Abstract	I
Table of Contents	III
List of Tables	V
1 Introduction	1
1.1 Motivation	1
1.2 Background	2
1.3 Outline	4
2 Literature Review	5
2.1 Literature Survey	5
2.2 Gaps in the Existing Literature	6
3 Model Principle	7
3.1 Principal Component Analysis Scoring Model (PCAS)	7
3.2 Granger Causality Test	10
3.3 Time Series ARIMA Model	12
4 Model Building and Solving	17
4.1 Model Assumptions	17
4.2 Data Preparation	18
4.3 Stocks Risk Scoring (PCAS Model)	22
4.4 Assess Market Sentiment	25
4.5 Predicting Market Sentiment(ARIMA Model)	28
5 Model Optimization	35
6 Conclusion and Future Work	36
6.1 Conclusion	36
6.2 Future Work	36
References	39
Appendix	42

List of Figures

Figure 4.1	SPSS Result	22
Figure 4.2	Risk scores for each stock in the communications equipment sector	25
Figure 4.3	Line chart of variance and closing price	27
Figure 4.4	Variance Time Series	29
Figure 4.5	Differenced Variance Time Series	30
Figure 4.6	ACF and PACF	31
Figure 4.7	SARIMAX Results	32
Figure 4.8	Improved ARIMA Forecast	34

List of Tables

Table 4.1	KMO and Bartlett's Test	23
Table 4.2	Component Matrix	23
Table 4.3	Coefficients in linear combinations	24
Table 4.4	Coefficient in the comprehensive score model	24
Table 4.5	Weights	24
Table 4.6	Parameter Estimates	33

1 Introduction

1.1 Motivation

In the recent period, global markets have been characterized by high volatility, with market sentiment being unpredictable. The reason for this change is influenced by many factors, especially major economic events, policy adjustments, wars, and global crises such as pandemics. For instance, the recent turmoil in the market has been at the mercy of the change in the U.S. interest rate cycle, the slowing economic growth in Europe, and uncertainties in the Chinese market. Meanwhile, the ongoing conflict between Russia and Ukraine, along with surging energy prices and global supply chain problems, has only increased market uncertainty. These have caused wild swings in investor sentiment to which market trends are susceptible. Investors are more likely to act irrationally in a high-volatility environment by panic selling or chasing stocks that surge. Therefore, understanding the impact of market sentiment on market volatility can help investors make more rational and sound decisions.

Similarly, market volatility might pave the way for investors to make more rational and logical decisions. Simultaneously, market sentiment is one of the leading indicators in projecting market trends. Analyzing sentiment for particular market sectors can apply more quantitative information to try to identify any up-and-coming changes in the market, therefore maximizing decision-making foresight and optimizing an overall investment strategy. Aside from the decisions to invest, market sentiment will also play a very important part in risk management.

Management in the context of rapid development, especially interdependence and complexity of financial markets, requires even more vital risk management. The systemic and individual stock risks can be identified by analyzing the relationship between the stock market sentiment and market risk. For example, some risk-scoring models would help investors analyze credit and liquidity risks in the market better and give quantitative support for decisions. It gets further deepened through time series analysis when monitoring historical market variance

data with a view to predicting the future development of market trends. These are complementary techniques: the risk-scoring model will give a snapshot of what the current state of the market risk is, while time series analysis provides the way to trace changes in sentiment fluctuations over time. This will be a means of ensuring that investors and regulators can identify any unusual market movement in good time and thereby take appropriate action to respond effectively.

1.2 Background

With the rapid development of information and communication technologies, the concept of a digital economy has gradually come into the view of the public and become the main factor that promotes economic and social development around the world. Especially in China, with the rise of the digital economy, the transformation of all walks of life has been further driven. Digital networks and robust communication infrastructure provide a platform for interaction and collaboration for individuals and organizations from around the world. In particular, the communications equipment industry is of great importance for the growth of the digital economy and industrial transformation. The economic activities in the United States, the United Kingdom, Australia, and Japan are gradually shifting from traditional offline modes to online platforms in promoting digital transformation [16]. This shift has been improving efficiency for economic activities, driving emerging digital industries to grow rapidly, and thus promoting economic enlargement. A great accomplishment was made by China in its current development of a digital economy. With a massive internet user base and the leading e-commerce market, China is bound to create certain advantages in driving digital growth. In the "14th Five-Year Plan," high-quality development has been stressed by China, focusing on the digital economy and promoting the digital, networked, and intelligent transformation of industries. To achieve this goal, the government has introduced policies such as the "General Layout Plan for Building Digital China" and the "14th Five-Year Digital Economy Development Plan," which aim to accelerate the construction of digital infrastructure,

promote technological innovation, cultivate digital industries, and further promote high-quality economic development [19]. According to the "2023 China Digital Economy Development Report" released by the China Academy of Information and Communications Technology, the scale of China's digital economy reached 50.2 trillion yuan in 2022, with a year-on-year growth rate of 10.3%, marking the 11th consecutive year of growth exceeding GDP growth [4]. This further underlines the importance of the digital economy in the economy of China. With the rapid development of the digital economy, risk analysis in financial management and investment decisions has become increasingly important. In particular, with new risk factors such as data leakage and cyber-attacks being increasingly prominent in the digital financial environment, traditional financial risk management methods need to innovate and improve. In this context, risk scoring, VaR (Value at Risk) Model, GARCH (Generalized Autoregressive Conditional Heteroskedasticity) Model, Monte Carlo Simulation and Risk Scoring Model have been widely applied, helping to quantify risks, capture market fluctuations, and assess risks [3]. In particular, sentiment analysis, as an emerging research field in financial markets, provides new perspectives on understanding market dynamics by analyzing investor sentiment, challenging the rational investor assumption in traditional financial theories. Combining this with the time series analysis means that the researcher can visualize the patterns of changes of sentiment over time and therefore serves as a far-reaching reference towards making an investment decision. Indeed, due to the revolution in the digital economy, fluctuations in the global market have led to unpredictability and increasingly complex emotional dynamics regarding China's communications equipment industry. The intricate dynamics of these will be delved into within the context of comprehensive analysis, interaction of market sentiment, and risk factors in the communications equipment sector, while further exploring how to scientifically make use of the work of sentiment analysis and risk management to provide more accurate investment decisions.

1.3 Outline

This specific study targets the Chinese A-share communication equipment industry, making decision-making references for investors by variance analysis of risk scores. The first step is data collection, which includes collecting data on publicly listed companies within the industry. At this step, several key risk indicators are used, such as the turnover rate, historical volatility, beta coefficient, Sharpe ratio, and Value at Risk, in a time-series format. Therefore, such multi-measure indicators of measurement in this case would help evaluate several dimensions of characteristics related to risks of the stock. These will ensure the consistency of data with the potential of comparability and provide the ground on which the analysis shall be laid. The stage following this data collection and organization shall be of the calculation of risk scoring with an aggregate score. Weightage of each indicator for derivation of the comprehensive risk score for every stock in the industry will be calculated and assigned through principal component analysis, which will enhance the accuracy of the scores. Subsequently, calculate the variance of the risk scores and further analyze this data through Granger causality tests to verify whether variance fluctuations can be used to predict changes in market sentiment. This is further explored by confirming the effectiveness of variance in reflecting changes in market sentiment and whether it is representative of market sentiment. Based on the results of the risk score variance analysis, insights will be provided into predicting market sentiment in the sector. Specifically, it will calculate the daily variance of risk scores, compile historical variance data, and systematically analyze and summarize trend changes through ARIMA time series analysis. This approach allows us to forecast future trends in market volatility for the communications equipment industry.

This would not only help in understanding the dynamic changes in market sentiment but will also function as a reference for investors to make more strategic decisions in such a complex market environment. This study adopts this systematic analytical framework to uncover potential investment opportunities and risks within the communications equipment industry and provide insights to the participants in the marketplace.

2 Literature Review

2.1 Literature Survey

In the field of financial market research, risk management, sentiment analysis, and time series forecasting are three major areas of focus. Extensive literature has explored these topics, proposing various methods and models, each with different applications and suitable contexts. Risk management plays a crucial role in bank credit assets, evolving from early experience-based judgment methods to modern risk assessment approaches that combine qualitative and quantitative analysis, advancing continually in response to technological and market changes. The development process of risk management can be roughly divided into several stages: **First Stage:** Experience-based Judgment Methods (before the 1970s) In the late 1950s, the renowned American consulting firm introduced the Delphi method, which used expert scoring to analyze qualitative indicators that were difficult to quantify, for the purpose of identifying risk. During this period, Western commercial banks primarily relied on the “5C” method to assess customers' credit risk based on experience, but lacked technology-supported risk assessment tools [7]. **Second Stage:** Combining Qualitative and Quantitative Analysis (1970s to late 1980s) Following the collapse of the Bretton Woods system, the global economic landscape changed, leading commercial banks to place greater emphasis on risk management and to incorporate quantitative analysis methods into their risk assessments [15]. **Third Stage:** Model Innovation (1990s to early 21st century) With the rapid development of the global economy and financial markets, commercial banks faced increased operational risks. In the early 1990s, J.P. Morgan and the “Group of 30” introduced the Value at Risk (VaR) method, whose comprehensive risk measurement capabilities made it a primary risk management tool for financial institutions. In 1993, KMV Corporation, based on the Black-Scholes-Merton (BSM) model and incorporating improvements from Longstaff and Schwarz as well as Dsa and Zhou, launched the KMV model, which evaluates risk using expected default probability as a core indicator [12]. **Fourth Stage:** Diversification

of Risk Assessment Models (early 21st century to present) In 2003, J.P. Morgan and other international banks introduced the first model for assessing credit risk in securitized portfolios, which calculates capital requirements for credit portfolios based on VaR principles. Starting in 2005, various consulting firms and banks developed new risk assessment models, such as AXIOM's Risk Monitor model and Algorithmic's Risk Watch model [3]. In 2009, British scholars Feng Yulin and Sally Mcclean combined four standalone early warning models—logistic regression, decision tree, neural network, and discriminant analysis—into multiple combination models and validated their effectiveness through empirical research [7].

Current Stage: Intelligent Risk Management Driven by Fintech and Big Data. With the rapid advancement of big data and fintech, risk management is transitioning towards intelligent solutions. According to Cao Yunchun et al. (2020), the application of fintech in supply chain finance can address traditional challenges in identifying supply chain relationships and correlating credit risk indicators through data mining and cross-industry collaboration [3]. Zhao Chengguo and Jiang Wenxin (2021), based on the logic of supply chain finance ecosystem development, constructed a risk management mechanism that integrates internal and external collaboration, and proposed recommendations for risk assessment and management [23].

2.2 Gaps in the Existing Literature

In recent years, the application of multimodal data processing technologies in risk management has gradually increased. Although these technologies can significantly improve the information collection process of traditional models, especially in integrating structured and unstructured data (such as text, images, etc.), their application in the financial sector is still in its early stages. There are still relatively few applications in financial scenarios such as intelligent risk control, customer marketing, and smart operations. Therefore, more empirical research is needed to explore how multimodal data can enhance the accuracy and efficiency

of financial risk assessment, as well as how these technologies can be applied to real financial businesses and decision-making.

At the same time, it is found that the existing literature has paid relatively little attention to the quantification of market sentiment, particularly in terms of empirical analysis within specific industries. Although there has been some research on market sentiment, most studies have not delved deeply into specific industries. The existing literature often lacks a discussion that combines industry backgrounds with practical application scenarios, and it has not focused in depth on fields with specific industry characteristics.

Moreover, while time series methods are widely used in predicting financial market sentiment, few studies have applied these methods to the quantitative analysis of market sentiment, especially in empirical analyses that correlate with actual market sentiment fluctuations.

In summary, this paper will absorb the advanced experience from existing literature, focusing on the integration of theoretical research and empirical analysis, and will concentrate on the Chinese A-shares communications equipment industry, providing a concrete industry background and application scenario. At the same time, ARIMA time series analysis aims to predict market sentiment fluctuations in the communications equipment industry, providing a new empirical case and effective quantitative analysis tools for time series forecasting and market sentiment analysis in financial market research. This research aims to provide new tools for risk management in this industry, as well as offer useful experience and methods for risk assessment research in other industries, thereby filling the gap in the literature on the quantification of market sentiment.

3 Model Principle

3.1 Principal Component Analysis Scoring Model (PCAS)

Principal Component Analysis (PCA) is a dimensionality reduction algorithm that extracts the most representative principal components from the data to reduce its dimensionality while preserving as much information as possible [14]. In financial risk assessment, PCA is often used to transform multiple related risk indicators into a few uncorrelated principal components, facilitating comprehensive evaluation and decision-making. In this scoring model, the primary purpose of using PCA is not dimensionality reduction, but rather to leverage the principles of PCA to calculate the weights of various risk indicators, which will assist in the subsequent risk scoring.

3.1.1 Data Standardization

Standardizing the data before performing PCA analysis is a crucial step. This is because different variables in the raw data may have different units of measurement and magnitudes, which can lead to certain variables dominating the analysis. After standardization, the mean of each variable is adjusted to 0, and the standard deviation is adjusted to 1, ensuring that all variables have equal importance in the analysis, thus eliminating the impact of differences in units of measurement.

The standardization formula is:

$$Z_i = \frac{X_i - \mu}{\sigma} \quad (3.1.1)$$

Where:

Z_i is the standardized value,

X_i is original value,

μ is mean,

σ is standard deviation.

3.1.2 Principal Component Analysis (PCA)

Calculate Covariance Matrix

Compute the covariance matrix C of the standardized data:

$$C = \frac{1}{n-1} Z^T Z \quad (3.1.2)$$

Where:

Z is the standardized data matrix,

n is the number of samples.

The covariance matrix reflects the linear relationships between the risk indicators.

Calculate Eigenvalues and Eigenvectors

Perform eigenvalue decomposition on the covariance matrix to obtain eigenvalues λ and eigenvectors v :

$$Cv = \lambda v \quad (3.1.3)$$

Eigenvalues indicate the amount of variance explained by each principal component, while eigenvectors indicate the direction of the principal components.

Calculate Indicator Weights

Use the eigenvalues and eigenvectors to calculate the weights of each indicator. The weight w_i can be determined using the corresponding eigenvalue:

$$w_i = \frac{\lambda_i}{\sum_{j=1}^p \lambda_j} \quad (3.1.4)$$

Where:

w_i is the weight of the i -th indicator,

λ_i is the corresponding eigenvalue,

p is the total number of indicators.

3.1.3 Scoring

Score based on the standardized indicators and their corresponding weights:

$$S = \sum_{i=1}^m w_i X_i \quad (3.1.5)$$

Where:

S is the overall score for the stock,

m is the number of risk indicators,

w_i is the weight of the i -th indicator,

X_i is the value of the i -th indicator.

The composite score derived from the scoring formula can reveal the overall risk level of each stock at a specific point in time.

3.2 Granger Causality Test

The Granger causality test is a method used to determine causal relationships between time series variables. It is based on the predictive ability of time series rather than a strict causal relationship. Proposed by Nobel Laureate in Economics, Clive Granger, this method is widely applied in fields such as financial markets and macroeconomics [11].

If, given the historical data, adding the historical data significantly improves the prediction, then is said to Granger-cause. Granger causality focuses on the predictive power of one variable's past information on another variable.

3.2.1 Stationarity Test

Before performing the causality test, it is essential to test the stationarity of the data, using methods such as the ADF (Augmented Dickey-Fuller) test or the KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test. If the data is non-stationary, differencing can be applied to achieve stationarity.

3.2.2 Model Building

Given two stationary time series and, based on the data of the two-time series, a VAR (Vector Autoregression) model and multiple lagged regression models are constructed. Specific steps include:

Univariate Autoregressive Model

Only past values of a single variable are used for regression analysis.

$$Y_t = \alpha_0 + \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \cdots + \alpha_p Y_{t-p} + \varepsilon_t \quad (3.2.6)$$

Multivariate Regression Model

Past values of an additional variable are incorporated into the regression analysis.

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots + \beta_p Y_{t-p} + \gamma_1 X_{t-1} + \gamma_2 X_{t-2} + \cdots + \gamma_q X_{t-q} + \eta_t \quad (3.2.7)$$

Appropriate lag orders p and q are selected, typically determined through information criteria such as AIC or BIC.

3.2.3 Model Testing

Hypothesis Testing

To determine if including the lagged terms significantly improves the prediction, conduct a hypothesis test with the following steps:

Null Hypothesis H_0 : X does not Granger-cause Y

$$\gamma_1 = \gamma_2 = \cdots = \gamma_q = 0 \quad (3.2.8)$$

Alternative Hypothesis H_1 : X Granger-causes Y

$$\text{at least one } \gamma_j \neq 0 \quad (3.2.9)$$

Statistical Test (F-Test)

The core of the Granger causality test lies in comparing two hypotheses. An F-test is conducted to determine whether the inclusion of past values significantly improves the

prediction. The null hypothesis of the F-test is that “ does not Granger-cause,” while the alternative hypothesis is that “ Granger-causes .” The formula for the F-test is:

$$F = \frac{(RSS_r - RSS_{ur})/q}{RSS_{ur}/(n - k)} \quad (3.2.10)$$

where:

RSS_{ur} represents the residual sum of squares from the univariate autoregressive model,

RSS_{ur} represents the residual sum of squares from the multivariate regression model,

q is the number of lagged terms introduced,

n is the sample size,

k is the number of explanatory variables in the model.

If the F-statistic is significantly greater than the critical value, or if the p-value is less than the significance level, the null hypothesis is rejected, indicating that X_t Granger-causes Y_t .

Bidirectional Testing

The Granger causality test is typically conducted bidirectionally. In addition to testing whether X Granger-causes Y, it is also essential to examine whether Y Granger-causes X. Bidirectional testing allows for a more comprehensive understanding of the causal relationship between the two variables.

3.3 Time Series ARIMA Model

ARIMA (AutoRegressive Integrated Moving Average) is a widely used statistical model for time series analysis. It is used to model and forecast time series data by capturing the autoregressive (AR), differencing (I), and moving average (MA) components. The ARIMA model is applicable to stationary time series data or data that can be transformed into stationary series through differencing [2].

The basic formula for the ARIMA model is:

$$Y_t = \mu + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t \quad (3.3.11)$$

Where:

Y_t is the observed value at time t ,

μ is the constant term (mean),

p is the order of the autoregressive (AR) part,

ϕ_i is the autoregressive coefficients, describing the relationship between the time series and its lagged values,

q is the order of the moving average (MA) part,

θ_j is the moving average coefficients, describing the relationship between the current value and past error terms,

ϵ_t is the white noise error term, representing the random component that the model cannot explain.

3.3.1 Data Stationarity

Before using the ARIMA model, it is important to ensure that the time series data is stationary. If the data exhibits trends or seasonality, differencing is necessary to remove these components and achieve stationarity.

3.3.1.1 Stationarity Test

The Augmented Dickey-Fuller (ADF) Test is a statistical test used to check for the presence of a unit root in a time series, which helps to determine whether a time series is stationary. It is an extension of the original Dickey-Fuller test and is capable of handling more general cases, especially when the series includes lagged terms [5].

Null Hypothesis and Alternative Hypothesis of the ADF Test

The ADF test is used to test the null hypothesis of the presence of a unit root. The hypotheses are as follows:

Null Hypothesis H_0 : The time series has a unit root (it is non-stationary).

Alternative Hypothesis H_1 : The time series does not have a unit root (it is stationary).

The goal of the ADF test is to determine whether it can reject the null hypothesis, thus confirming whether the series is stationary or not.

Basic Model of the ADF Test

The ADF test checks for unit roots using a regression equation based on the following model:

$$\Delta Y_t = \alpha + \beta t + \gamma Y_{t-1} + \sum_{i=1}^p \delta_i \Delta Y_{t-i} + \epsilon_t \quad (3.3.12)$$

Where:

Y_t is the observed value of the time series,

$\Delta Y_t = Y_t - Y_{t-1}$ is the first difference of the time series,

α is a constant term (sometimes with a trend term βt),

γ is the coefficient of the lagged value Y_{t-1} , which is crucial for testing the unit root,

δ_i are the coefficients of the lagged differenced terms, introduced to account for serial correlation in the data,

ϵ_t is the error term (assumed to be white noise).

The key parameter in this equation is γ . If $\gamma = 0$, this indicates the presence of a unit root, suggesting the series is non-stationary. If $\gamma < 0$, it indicates the absence of a unit root, meaning the series is stationary.

Steps of the ADF Test

Construct the regression model: Based on the time series data, construct the above regression model, selecting an appropriate lag length p .

Estimate the regression model: Use ordinary least squares (OLS) to estimate the model parameters and calculate the test statistic.

Test the significance of γ : Compute the t-statistic for γ and compare it with critical values.

Interpretation of the ADF Test Result

p-value < 0.05 : Reject the null hypothesis, implying the time series is stationary.

p-value $\geq$ 0.05: Fail to reject the null hypothesis, implying the time series has a unit root and is non-stationary.

3.3.1.2 Differencing

If the data is not stationary, differencing is applied to achieve stationarity.

The differencing formula is:

$$\Delta Y_t = Y_t - Y_{t-1} \quad (3.3.13)$$

If the data remains non-stationary after first differencing, second differencing or higher-order differencing can be applied. Each differencing operation increases the differencing order d .

3.3.2 Determining the ARIMA Model Order

When constructing an ARIMA model, it is crucial to determine the appropriate orders: p (autoregressive part order), d (differencing order), and q (moving average part order). The order determination is typically done using the ACF (Autocorrelation Function) and PACF (Partial Autocorrelation Function) plots.

Order of the Autoregressive (AR) Part p

The AR part describes the relationship between the current value and past values. The order p can be determined by observing the PACF plot. The PACF plot exhibits a sharp drop and becomes insignificant after the p -th lag when the order is correct.

Order of the Moving Average (MA) Part q

The MA part uses past error terms to adjust the current prediction. The order q can be determined by observing the ACF plot. The ACF plot exhibits a sharp drop and becomes insignificant after the q -th lag when the order is correct.

Differencing Order d

The differencing order d refers to the number of times the data must be differenced to achieve stationarity. If the data is stationary, $d = 0$; if one differencing is required, $d = 1$, and so on.

Comprehensive Order Selection

When selecting the model orders, information criteria like AIC (Akaike Information Criterion) or BIC (Bayesian Information Criterion) are used to balance the model's fit and complexity, ensuring the optimal choice of orders.

3.3.3 ARIMA Model Fitting and Parameter Estimation

Once the appropriate orders (p, d, q) have been determined, statistical methods can be used to fit the ARIMA model. Common estimation methods include Maximum Likelihood Estimation (MLE) and Ordinary Least Squares (OLS). These parameters can be estimated using statistical software such as R or the Statsmodels library in Python.

By fitting the model to actual data, the autoregressive coefficients ϕ_i and moving average coefficients θ_j can be obtained.

3.3.4 Model Diagnostics and Residual Checks

After fitting the ARIMA model, diagnostics must be performed to validate the model's suitability. The key diagnostic is the behavior of the residuals (error terms). Ideally, the residuals should be zero mean, uncorrelated, and normally distributed white noise.

Residual Autocorrelation Test

The Ljung-Box test is commonly used to test for residual autocorrelation. If the p-value is greater than the significance level (usually 0.05), the residuals are considered to have no significant autocorrelation, indicating that the model is well-fitted.

Residual Analysis

ACF and PACF plots of the residuals can be drawn to check for significant autocorrelation. If the residual plots show random behavior with no clear pattern, the model is well-fitted. If there is significant autocorrelation, adjustments to the model, such as changing the AR or MA order, may be needed.

3.3.5 Model Forecasting

Once the ARIMA model has been fitted and validated, it can be used for forecasting.

The ARIMA forecasting formula is:

$$\hat{Y}_{t+h} = \mu + \sum_{i=1}^p \varphi_i Y_{t+h-i} + \sum_{j=1}^q \theta_j \epsilon_{t+h-j} + \epsilon_{t+h} \quad (3.3.14)$$

Where:

$\hat{Y}_{t+h}$ is the forecasted value,

h is the forecast horizon (i.e., the number of steps ahead),

p, q, μ are the parameters obtained from fitting the model.

The ARIMA model can be used to generate predictions for future time points, based on historical data and model parameters. The forecast accuracy can then be evaluated by comparing predicted values with actual values from a holdout dataset.

4 Model Building and Solving

4.1 Model Assumptions

Hypothesis 1: Correlation Among Risk Indicators

It is assumed that there is a significant linear relationship among different risk indicators (such as turnover rate, volatility, Beta, Sharpe ratio, etc.). By conducting covariance analysis, these indicators can be examined to understand how they influence each other, thereby affecting the overall market risk level.

Hypothesis 2: Consistent Impact of Risk Indicators on Stock Performance

It is assumed that under similar market conditions, the impact of different risk indicators on price fluctuations and returns is similar across various stocks. This implies that the risk characteristics of different stocks can be described using the same risk model, thus improving the model's generalizability.

Hypothesis 3: Correlation Between Market Sentiment and Risk Score Variance

It is assumed that controlling for other factors, there is a significant correlation between the variance of risk scores and market sentiment. For example, higher variance in risk scores may be accompanied by more pessimistic market sentiment, as reflected in sentiment analysis results.

Hypothesis 4: Stability of Model Parameters

It is assumed that the parameters used in the model (such as risk weights, model coefficients, etc.) remain stable throughout the study period. This suggests that changes in the external economic environment will not significantly affect model parameters, thereby ensuring the model's predictive ability.

Hypothesis 5: Symmetry in Market Reactions

It is assumed that the market's reaction to changes in risk is symmetric. When risk increases, market sentiment tends to become pessimistic; conversely, when risk decreases, market sentiment tends to become optimistic. This hypothesis can be validated by examining the relationship between sentiment data and risk scores.

Hypothesis 6: Stability of Market Structure

It is assumed that there are no major structural changes in the market during the study period, such as policy shifts, technological innovations, or sudden market behavior.

4.2 Data Preparation

4.2.1 Data Collection and Cleaning

This study selected the communications equipment industry from the Wind stock sector screening column and gathered data on six key metrics — turnover rate, volatility, beta coefficient, Sharpe ratio, and Value at Risk (VaR) — for 173 listed companies in this industry from August 19, 2024, to November 15, 2024. This data covers a total of 58 days of market performance.

Turnover Rate

The stock turnover rate is a critical index reflecting the liquidity of the market, and the frequency of trading. On one hand, if the turnover rate is higher, the stock market tends to be more dynamic; otherwise, when the turnover rate is low, the situation becomes less dramatic. The daily turnover rate from August 19 to November 15, 2024, can be gathered to observe changes in market liquidity and sentiment fluctuations during this period. These data reflect short-term market volatility, especially in those periods when there was an unprecedented price movement, and during those times, the turnover rate usually indicates changes in investor sentiment, such as panic or speculative buying. For example, if a sudden increase in the rate of a stock or industry turnover is noticed, it always signals the shift in investor appetite for risk, caused by changes in market expectations regarding this stock or sector. Stocks with high turnover rates might have wild short-term oscillations in prices, which becomes very important for investors to decide on their buying or selling activities.

Volatility

Volatility reflects the extent of price fluctuations in stocks and is usually measured by standard deviation. The higher the volatility, the more uncertainty exists in the market and the higher the investment risk. Conversely, lower volatility suggests a more stable market with reduced risk. This study will be collecting daily volatility data from August 19 to November 15, 2024, to aid in analyzing short-term market fluctuations. Volatility changes are of particular concern for any short-term investor, as they directly influence the dispersion of investment returns. High volatility usually goes hand in hand with high risk, which is a

crucial factor in investment decisions. By observing volatility changes, points of sharp changes in market sentiment can be detected, especially when market fluctuation suddenly heightens, which may indicate extraordinary investors' emotions, such as panic sell-off or speculative purchase. Regular monitoring of volatility allows investors to adopt measures of risk avoidance in a timely manner or seize opportunities presented by the market.

Beta Coefficient

The beta coefficient expresses a stock's volatility regarding the general market. A beta of 1 indicates that the stock moves in line with the market, while a beta above 1 signifies that the stock is more volatile than the market, and a beta below 1 indicates lower volatility compared to the market. It will gather daily beta coefficient data from August 19 to November 15, 2024, to analyze the systematic risk of the stocks in the communication equipment industry. The beta coefficient helps investors understand a stock's sensitivity to market fluctuations and adjust their portfolio's risk exposure accordingly. For example, a higher beta means that the stock gives higher returns when the market rises but also carries greater risks when the market falls. Therefore, investors can adjust their investment strategies based on beta coefficients, such as diversifying to reduce risk from high-beta stocks or choosing low-beta stocks to reduce volatility risk.

Sharpe Ratio

The Sharpe ratio is a measure for assessing risk-adjusted returns by calculating the ratio of excess return, i.e., asset return minus the risk-free return, to standard deviation. A higher Sharpe ratio indicates that, on the same amount of risk taken, the investor is earning more significant returns. It will collect daily Sharpe ratio data starting from August 19 to November 15, 2024. This metric proves very effective in analyzing the stock or a portfolio's risk-return profile in high-volatility market scenarios. In general, the Sharpe Ratio assists investors in choosing those investment options that promise higher returns while keeping their risks under control. In volatile market conditions, the Sharpe ratio will help investors

identify which stocks or sectors are better in terms of risk management. In combination with other indicators, such as volatility and turnover rate, it will enable investors to make more rational and forward-looking investment decisions.

Value at Risk (VaR)

Value at Risk is a means of quantifying the maximum potential loss of an asset. It usually stands for the maximum probable loss at a certain confidence level over a certain period. With VaR, investors are able to gauge the downside risk of a stock or portfolio under extreme market conditions. It will collect daily VaR data from August 19 to November 15, 2024, to identify potential risks in the communications equipment industry, especially during events such as policy changes or global crises. The heart of the VaR model is the estimation of the maximum expected loss through a set level of confidence, such as 95% or 99%, to showcase the probable loss an investor would incur in the worst situation. For instance, with a VaR of 1%, it indicates that, with a probability of 95%, the highest loss the following day will not be more than 1% of the stock's total market value. Investors can use the VaR value to gauge the risk of their portfolio and make corresponding hedging actions.

Handling missing values and outliers, to ensure the integrity and accuracy of the data.

4.2.2 Construct Data Matrix

Organize the risk indicator data into a matrix X , where each row represents a stock and each column represents a risk indicator.

4.2.3 Data Standardization

Standardize the data X_{ij}

$$Z_{ij} = \frac{X_{ij} - \mu_j}{\sigma_j} \quad (4.2.1)$$

Where:

Z_{ij} is the standardized value of the j -th indicator for the i -th stock,

X_{ij} is the original value,

μ_j is mean of the j -th indicator,

σ_j is standard deviation of the j -th indicator.

4.3 Stocks Risk Scoring (PCAS Model)

4.3.1 Model Objective

Evaluating the risks of selected stocks and generating comprehensive risk scores, to provide data support for the subsequent judgment of sector sentiment.

4.3.2 Model Establishment

Use data from August 19, 2024 as an example.

Use Principal Component Analysis to Calculate Indicator Weights

Use the Factor analysis function (PCA) in SPSS to obtain the Result 1 as follows. SPSS Operation Procedures refer to Appendix A.

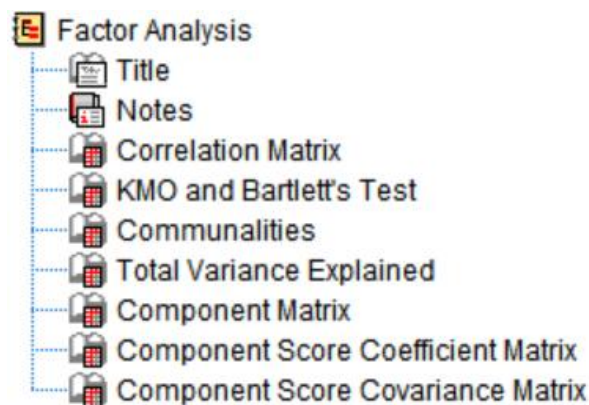


Figure 4.1 SPSS Result

Testing

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		0.472
Bartlett's Test of Sphericity	Approx. Chi-Square	1387.790
	df	15
	Sig.	< 0.001

Table 4.1 KMO and Bartlett's Test

The results show that the KMO statistical value is greater than 0.5, and it can be seen that the degree of correlation between variables is not much different; The result of Bartlett's sphericity test is less than 0.05, the sphericity hypothesis is rejected, and there is a correlation between the original variables, so the data is suitable for factor analysis.

Weights Calculation

Use the data in Result 1 to calculate the coefficients in linear combinations.

Component Matrix		
	Component 1	Component 2
Zscore: Volatility	0.9560 a_1	-0.0084
Zscore: VaR	0.9502 a_2	-0.0145
Zscore: Sharpe	0.7348 a_3	0.3145
Zscore: Beta	-0.3693 a_4	0.0970
Zscore: Turnover	-0.0462 a_5	0.8736
Zscore: Swing	-0.1573 a_6	0.8462
Component Eigenvalue	2.5200 α	1.5877

Table 4.2 Component Matrix

The following table is calculated by the formul:

$$\text{Coefficients in linear combinations} = a_i / \alpha \quad (4.3.2)$$

Coefficients in linear combinations		
	Component 1	Component 2
Zscore: Volatility	0.6022 x_1	-0.0067 y_1
Zscore: VaR	0.5986 x_2	-0.0115 y_2
Zscore: Sharpe	0.4629 x_3	0.2496 y_3
Zscore: Beta	-0.2326 x_4	0.0770 y_4

Zscore: Turnover	-0.0291 x_5	0.6933 y_5
Zscore: Swing	-0.0991 x_6	0.6715 y_6
Component Variance	42.0004 x	26.4625 y

Table 4.3 Coefficients in linear combinations

Then, calculate the coefficient in the comprehensive score model.

$$\text{Coefficients in the comprehensive score model} = \frac{x_i \cdot x + y_i \cdot y}{x + y} \quad (4.3.3)$$

Coefficient in the comprehensive score model	
Zscore: Volatility	0.3669 h_1
Zscore: VaR	0.3628 h_2
Zscore: Sharpe	0.3804 h_3
Zscore: Beta	-0.1130 h_4
Zscore: Turnover	0.2501 h_5
Zscore: Swing	0.1988 h_6

Table 4.4 Coefficient in the comprehensive score model

Finally, calculate the weights.

$$w_i = \frac{h_i}{\sum h_i} \quad (4.3.4)$$

Weights	
Zscore: Turnover	0.2537
Zscore: Swing	0.2509
Zscore: Volatility	0.2631
Zscore: Beta	-0.0781
Zscore: Sharpe	0.1730
Zscore: VaR	0.1375

Table 4.5 Weights

Calculate the Risk Scoring

Score based on the standardized indicators and their corresponding weights:

$$S = \sum_{i=1}^m w_i X_i \quad (4.3.5)$$

Where:

S is the overall score for the stock,

m is the number of risk indicators,

w_i is the weight of the i-th indicator,

X_i is the value of the i-th indicator.

4.3.3 Model Results

After calculation, the risk scores of all the stocks in the communication equipment sector at a certain time is obtained.

The visual rating results are as follows.

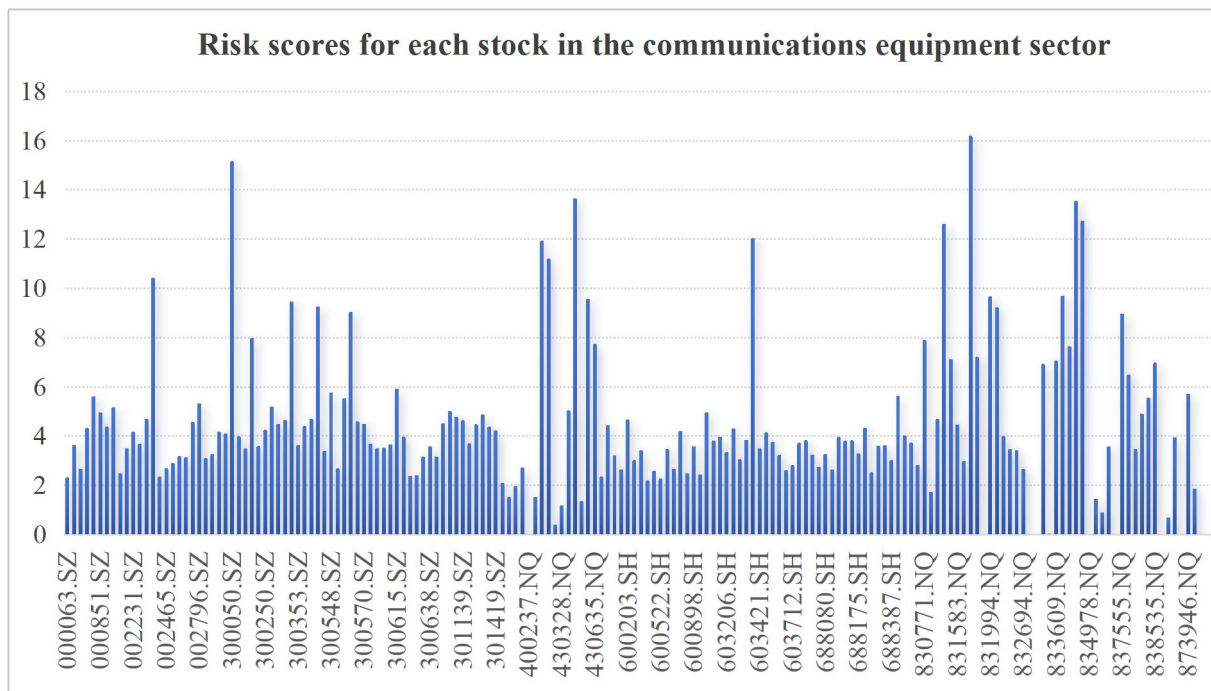


Figure 4.2 Risk scores for each stock in the communications equipment sector as of August 19, 2024

4.4 Assess Market Sentiment

4.4.1 Mean Calculation

Obtain the average risk score of all stocks in the sector to provide an overview of the overall risk level. Use excel to calculate the mean of risk scores is 4.388.

4.4.2 Variance Calculation

Calculate the variance of risk scores to assess the degree of dispersion. Use excel to calculate the variance of risk scores is 8.5761.

4.4.3 Assess Market Sentiment (Granger Causality Test)

Model Objective

The objective is to determine whether the variance of the risk score can be used to reflect market sentiment.

Model Establishment

Data Description

The dataset used includes two main time series (2024-08-19 to 2024-11-15):

Closing Price (P_t): The daily closing price of the communication equipment sector.

Risk Score Variance (V_t): The variance of risk score at each time point is calculated according to the stock risk score obtained in 5.3.2.

Stationarity Check

Granger causality testing requires stationary data. Using the Augmented Dickey-Fuller (ADF) test to check for stationarity in both P_t and V_t .

	t	P	critical value		
			1%	5%	10%
V_t	-0.469	0.898	-3.555	-2.916	-2.596
P_t	-0.644	0.861	-3.553	-2.915	-2.595

The P-values of P_t and V_t are all larger than 0.05. The null hypothesis cannot be rejected, and the series are not stationary time series.

Data Stationarity

Perform differencing on V_t and P_t respectively to achieve data stationarity.

Model Solution

The test results of Granger causality obtained by using SPSS are as follows

		F	P
P_t	V_t	3.78	0.030
V_t	P_t	6.78	0.002

Based on the closing price and variance of the variable, the significance P-value is 0.030, which shows significance. Null hypothesis can be rejected, and the closing price can cause variance changes.

Based on the variance of the variable and the closing price, the significance P value is 0.002, showing significance. Rejecting the null hypothesis, the variance can cause the rate of return to change.

Model Result

The Granger causality test provides insights into the predictive relationship between closing prices and risk score variance in the communications equipment industry. The test results indicate that the risk score variance Granger-causes the closing price, suggesting that fluctuations in the risk score variance can serve as an indicator of changes in market sentiment. It provides a basis for using the risk score variance as a predictive tool for market dynamics in the communications equipment industry. The significance of the P-values underscores the reliability of this causal relationship.

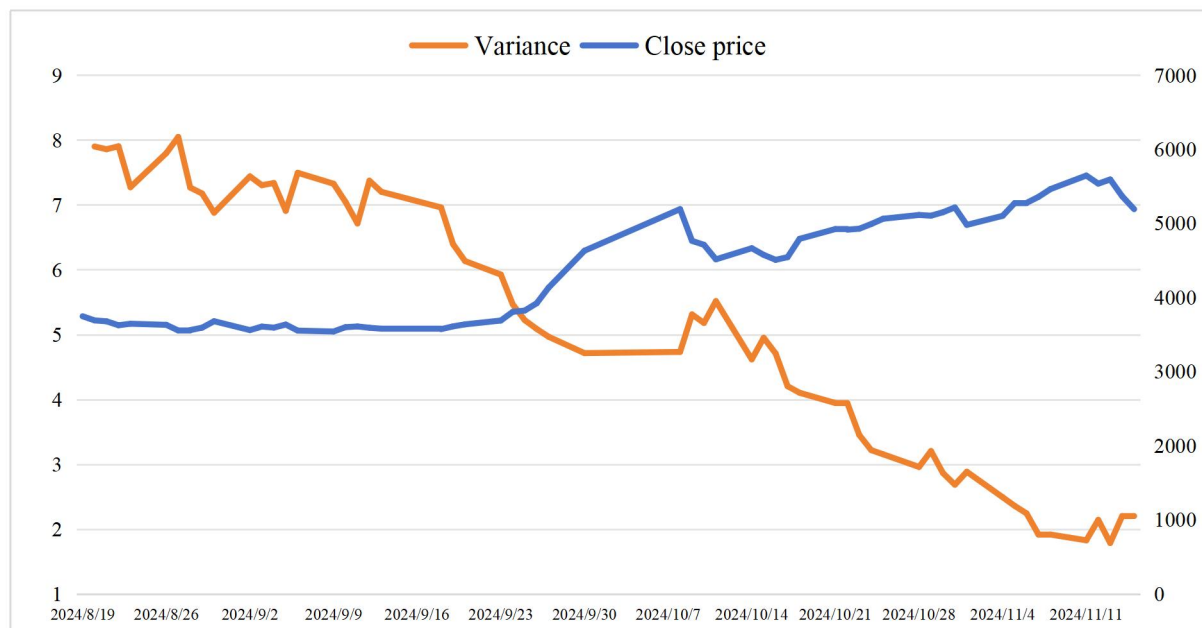


Figure 4.3 Line chart of variance and closing price

4.4.4 Variance Reflects Market Sentiment

By assessing the variance of stock risk scores, investors can effectively glean clues about market sentiment. The level of variance not only reflects differing expectations of risk among market participants but also reveals the stability and volatility of market sentiment.

High variance indicates significant differences in stock risk scores

Investors have inconsistent expectations regarding future trends, leading to unstable market sentiment. Insufficient confidence among investors may result in drastic price fluctuations or even declines.

Low variance signifies concentrated stock risk scores

Investors share a consistent understanding of market or sector risks, contributing to more stable market sentiment. Higher market confidence and aligned investor perspectives create an optimistic sentiment, potentially driving stock prices steadily upward.

4.5 Predicting Market Sentiment(ARIMA Model)

The ARIMA (Auto Regressive Integrated Moving Average) model is widely used for time series forecasting, including predicting market sentiment based on historical risk scores. This section discusses the process of building an ARIMA model to forecast market sentiment, using variance data of risk scores as the input.

4.5.1 Data Preprocessing and Visualization

The variance of the risk scores and their corresponding time indices were computed and stored in the Variance.csv file. The data spans from 2024-08-19 to 2024-11-15.

Reading Data

The risk score variance data is read from the Variance.csv file into a DataFrame for further processing.

Visualization

A time series plot is created to observe the trend, seasonality, and volatility of the risk score variance over the specified period.

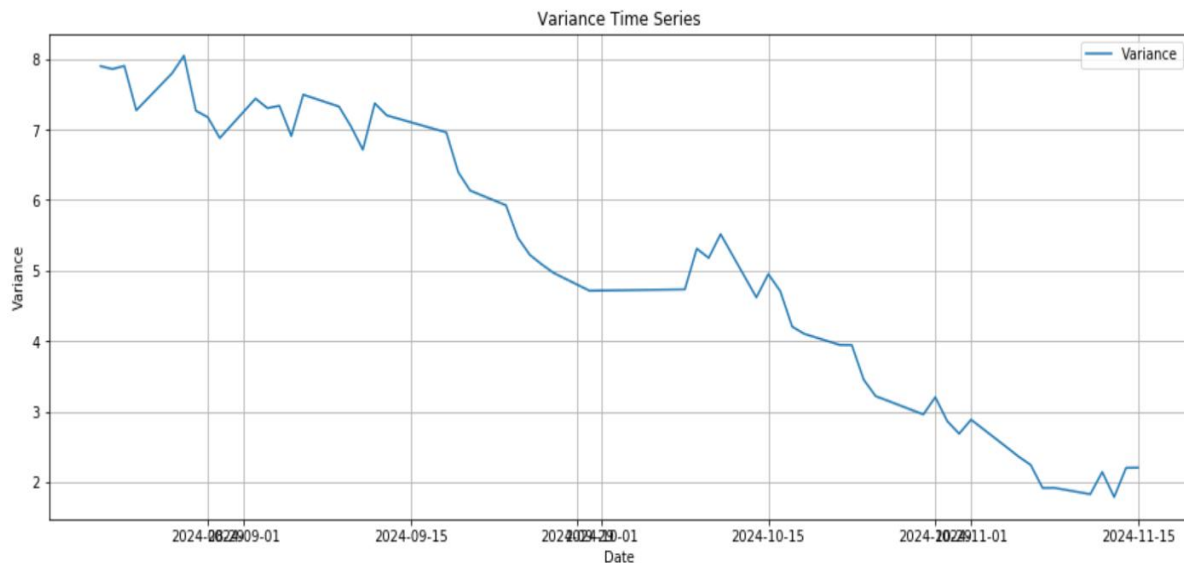


Figure 4.4 Variance Time Series

4.5.2 Data Stationarity

To apply ARIMA, the data needs to be stationary. A stationary time series has constant mean, variance, and autocorrelation over time.

Stationarity Test (Augmented Dickey-Fuller Test)

The Augmented Dickey-Fuller (ADF) test is performed to check for stationarity.

Code:

```
# Perform ADF test
adf_test = adfuller(data['Variance'])

# Output ADF test results
print('ADF Statistic:', adf_test[0])
print('p-value:', adf_test[1])
print('Critical Values:', adf_test[4])
```

The test results:

ADF Statistic: -0.46879

p-value: 0.89796

Since the p-value is greater than the significance level (0.05), it is concluded that the data is non-stationary and needs differencing.

Differencing

The first-order difference is applied to the data to make it stationary. After differencing, it again conducts the ADF test: ADF Statistic: -9.90803 , p-value: $3.22432e - 17$.

Since the p-value is less than 0.05, it concludes that the data is now stationary.

Plotting the time series of the differenced risk score variance.

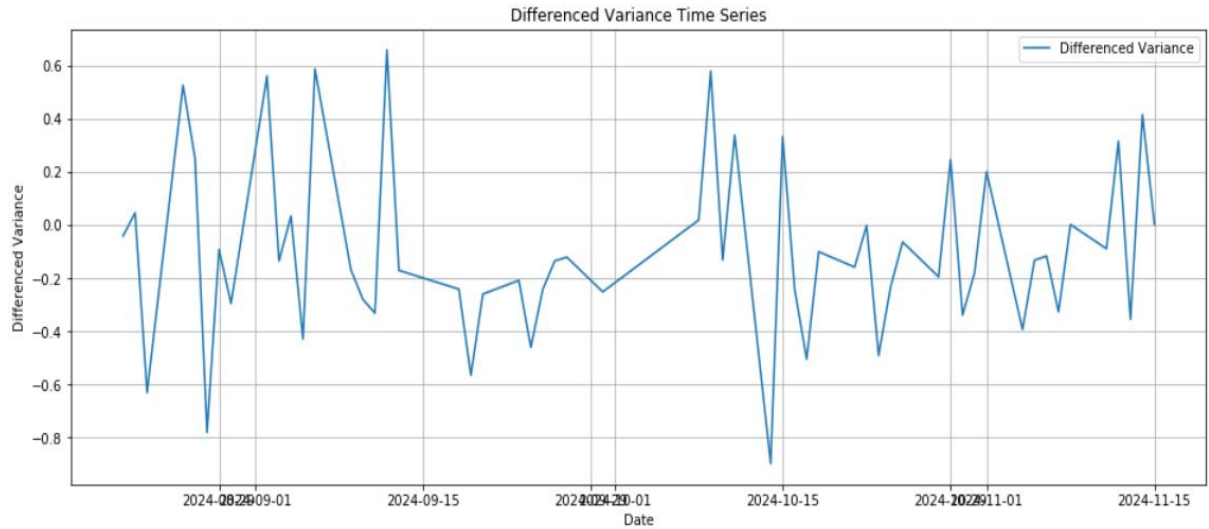


Figure 4.5 Differenced Variance Time Series

4.5.3 Building the ARIMA Mode

Once the data is stationary, the ARIMA model can be fitted.

The ARIMA model:

$$Y_t = \mu + \sum_{i=1}^p \varphi_i Y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t \quad (4.5.6)$$

4.5.4 ACF and PACF Plots

Plotting the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) to help determine the appropriate values for p and q .

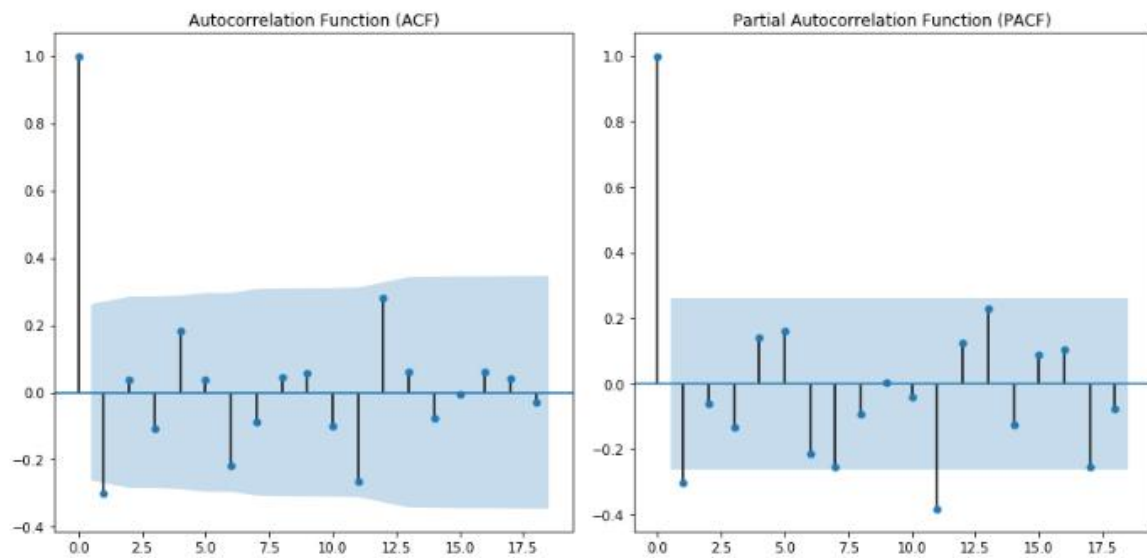


Figure 4.6 ACF and PACF

4.5.5 Data Split: Training Set and Test Set

Splitting the data into a training set (80%) and a test set (20%). The training set is used to build the model, while the test set is used for model validation.

Code:

```
# Split data into training set and test set (80% training set, 20% test set)
train_data, test_data = train_test_split(data, test_size=0.2, shuffle=False)
```

4.5.6 Parameter Selection and Model Training

Using the `auto_arima` function from the `forecast` package to automatically determine the best ARIMA model parameters. The function tries various combinations of p , d and q values to find the optimal model.

Code:

```

from pmdarima import auto_arima

# Automatically select the best ARIMA parameters based on AIC or BIC
auto_model = auto_arima(train_data['Variance'], seasonal=False, trace=True,
                        error_action='ignore', suppress_warnings=True, stepwise=True)

# Fit the best ARIMA model
model_fit = auto_model.fit(train_data['Variance'])

# Print the model summary
print(model_fit.summary())

```

Result:

```

Best model: ARIMA(0, 1, 1)(0, 0, 0)[0] intercept
Total fit time: 0.677 seconds

```

SARIMAX Results						
Dep. Variable:	y	No. Observations:	45			
Model:	SARIMAX(0, 1, 1)	Log Likelihood	-14.534			
Date:	Sun, 01 Dec 2024	AIC	35.067			
Time:	13:53:30	BIC	40.420			
Sample:	0	HQIC	37.052			
	- 45					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
intercept	-0.1143	0.039	-2.911	0.004	-0.191	-0.037
ma.L1	-0.3291	0.163	-2.014	0.044	-0.649	-0.009
sigma2	0.1131	0.035	3.222	0.001	0.044	0.182
Ljung-Box (L1) (Q):		0.00	Jarque-Bera (JB):	2.11		
Prob(Q):		0.99	Prob(JB):	0.35		
Heteroskedasticity (H):		0.63	Skew:	0.47		
Prob(H) (two-sided):		0.39	Kurtosis:	2.47		

Figure 4.7 SARIMAX Results

The equation of the model:

$$Y_t = -0.11425446 + \epsilon_t + 0.32907953\epsilon_{t-1} \quad (4.5.7)$$

4.5.7 Model Analysis

Parameter Estimates

Parameter	Estimate	Std. Error	z-value	p-value	95% Confidence Interval
Intercept	-0.1143	0.039	-2.911	0.004	[-0.191, -0.037]

Parameter	Estimate	Std. Error	z-value	p-value	95% Confidence Interval
Moving Average	-0.3291	0.163	-2.014	0.044	[-0.649, -0.009]
Residual Variance	0.1131	0.035	3.222	0.001	[0.044, 0.182]

Table 4.6 Parameter Estimates

Model Diagnostics

Ljung-Box Test: $Q = 0.00$, $p = 0.99$ (No significant autocorrelation)

Jarque-Bera Test: $JB = 2.11$, $p = 0.35$ (Residuals approximately normally distributed)

Heteroscedasticity Test: $H = 0.63$, $p = 0.39$ (No significant heteroscedasticity)

Skewness: 0.47 (Nearly symmetric)

Kurtosis: 2.47 (Close to normal distribution)

Information Criteria

AIC = 35.067 BIC = 40.420 HQIC = 37.052

Summary

This model is an ARIMA(0,1,1), fitting the data with a first-order differencing ($d=1$) and a first-order moving average (MA1). Both the intercept and the moving average term are significant, and the residuals show no autocorrelation, no heteroscedasticity, and are approximately normally distributed. The information criteria values suggest that the model fits the data well.

4.5.8 Forecasting and Result Visualization

Once the ARIMA model is trained, it is used to forecast the test set and future values. The forecasts are compared with the actual values to evaluate the model's performance.

Code:

```

# Make predictions for the test set and the next five steps
forecast_steps = len(test_data) + 5
forecast = model_fit.predict(n_periods=forecast_steps)

# Create forecast index
forecast_index = pd.date_range(start=test_data.index[0], periods=forecast_steps,
                               freq='B')

# Plot the comparison between actual and forecasted values
plt.plot(train_data.index, train_data['Variance'], label='Train Data')
plt.plot(test_data.index, test_data['Variance'], label='Test Data')
plt.plot(forecast_index, forecast, label='Forecast', color='red')
plt.title('Improved ARIMA Forecast')
plt.xlabel('Date')
plt.ylabel('Variance')
plt.show()

```

Result:

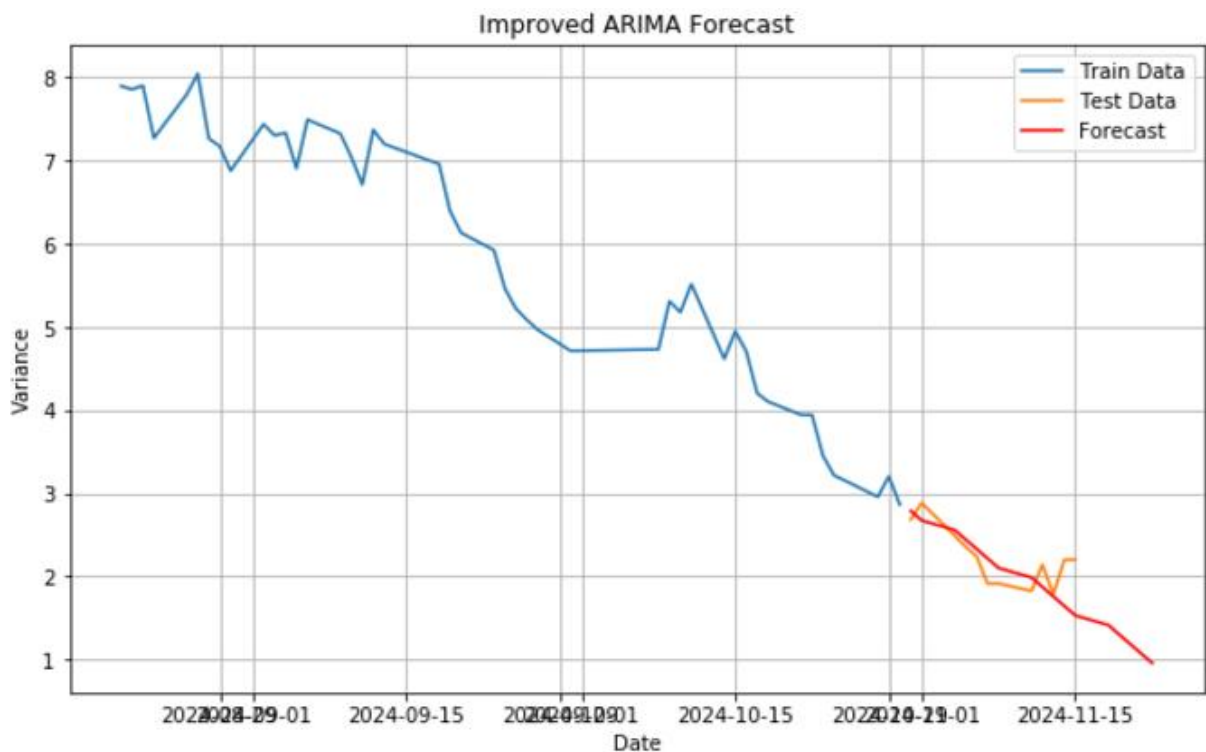


Figure 4.8 Improved ARIMA Forecast

Analysis:

In the test phase, the prediction of the model (red line) is close to the actual test data (orange line), which indicates that the model performs better on the test set and can capture the variation trend of the variance more accurately. In the forecasting stage, the model predicts that the variance will continue to decline, and the downward trend is relatively stable.

Therefore, this study can conclude that market sentiment in the sector will continue to rise, and investors have a consistent understanding of market or industry risks, which contributes to more stable market sentiment. Higher market confidence and consistent investor views have created optimism, driving stock prices steadily higher.

5 Model Optimization

In this study, this study combines Principal Component Analysis Scoring Model (PCAS), Granger Causality Test, and ARIMA model to explore the relationship between the risk and market sentiment fluctuations of stocks in the communications equipment industry. The model calculates weights and generates risk scores and their variance using PCA, verifies the causal relationship between variance and market sentiment through Granger Causality Test, and performs time series analysis on the variance using ARIMA to predict future market sentiment changes. This integrated model is more accurate than a single model, with a clear framework and highly operable steps, providing an effective risk warning system and decision support tool to help investors make more rational and robust investment decisions. However, each method has its advantages and limitations. Through an in-depth analysis of the existing modeling approaches, some optimization suggestions can be made to further enhance the overall model's predictive ability and reliability.

Future work can incorporate a series of strategies to improve the model's predictive accuracy, adaptability, and interpretability:

The Granger Causality Test, while effective in revealing causal relationships between variables, is limited to linear relationships. If nonlinear causal relationships exist in the data, an extension to nonlinear Granger Causality Testing could be considered, such as methods based on neural networks or support vector machines, to reveal more complex causal chains.

In ARIMA time series analysis, although the model is widely used for forecasting, it requires strong data stationarity and does not account for external factors. ARIMA can be combined with machine learning methods (such as Random Forest, XGBoost, etc.) to form a hybrid

model, improving the recognition of nonlinear relationships and increasing prediction accuracy. At the same time, the method for making data stationary can be improved by incorporating volatility modeling (such as GARCH models) to more effectively handle time series data with volatility clustering.

Given the dynamic nature of financial market data, it is recommended to build a real-time updating modeling framework that automatically adapts to new data, ensuring the model remains timely. Additionally, incorporating machine learning and deep learning interpretability techniques (such as LIME, SHAP) can improve model transparency, helping researchers and investors better understand the prediction mechanisms and enhancing the model's operability and trustworthiness. Through these optimizations, the model's predictive ability, adaptability, and transparency will be significantly improved, providing more accurate and robust support for investment decisions.

6 Conclusion and Future Work

6.1 Conclusion

This study focuses on the financial analysis of the communications equipment industry in the China A-share market and examines the relationship between market sentiment and risk assessment. The study collects and analyses various risk indicators (volatility, risk value, Sharpe, beta, sales and volatility) for all stocks in the industry and uses principal component analysis (PCA) risk rating, Granger causality testing and ARIMA time series forecasting models to data-based market sentiment forecasting system.

The study first standardises data on risk so that data from different scales and dimensions can be compared and analysed under the same criteria. Once standardised, the weights of the different risk indicators are calculated using PCA and the overall risk score for each stock in the industry is calculated based on these weights. This process takes into account several risk

indicators and the resulting risk score more comprehensively reflects the risk level of each stock.

To obtain time series data, the risk scores of all stocks are calculated for each point in time and the variance of the scores is determined. Assume that the differences reflect changes in market sentiment to some extent. Large differences mean that there is a significant discrepancy in the market's view of a stock's risk, which may cause the stock's price to fall. On the other hand, a small difference means that views on risk are more aligned, the market is more aggressive and the share price is higher. To investigate this relationship between variance and changes in share price, Granger's causality test was used to analyse the relationship between variance in risk score and changes in share price. The results confirm that the variance in risk scores (market conditions) reflects the change in share prices.

Finally, this study models the ARIMA model using time series data on the counterparty differences. The model predicts future trends in variance based on historical data and predicts future trends in market sentiment. The ARIMA model predictions show that market sentiment is predictable. This provides a powerful tool for investors to respond more appropriately to market fluctuations and adjust their investment strategies in a timely manner. In summary, the study reveals a significant causal relationship between the variance of risk scores analysed using multiple risk indicators and market volatility, and this relationship can be analysed and predicted quantitatively. This study provides investors and professionals in the financial risk management field with a new perspective and considers the volatility of market sentiment to assist in quantifying and managing risk in decision-making. However, the assessment of market sentiment and risk is complex and multidimensional, and the methods used in this study are limited and cannot fully capture all market movements. Future research can further explore and improve upon this foundation.

6.2 Future Work

This study explores the relationship between market sentiment and risk analysis in the Chinese A-share market, specifically within the communications equipment industry. Future research can build upon this foundation for further exploration and deepening.

Future research could adjust the data selection process. By applying the model in different market environments, the comprehensiveness of market sentiment and risk analysis can be enhanced. Specifically, the research can expand the scope to include international markets or further refine the data to focus on specific subfields within the communications equipment industry, such as 5G infrastructure and the Internet of Things (IoT). Analyzing data from different sectors will help better understand and address the specific risks and market changes in each area, thus aiding investors and decision-makers in making optimal decisions in a volatile market environment.

When calculating risk scores, in addition to the five indicators chosen in this study, more risk indicators, such as liquidity risk and credit risk, can be considered. Furthermore, by incorporating machine learning models, a non-linear risk scoring model can be developed, making risk assessment more comprehensive and accurate.

In the analysis of market sentiment, additional factors could be introduced, especially policy shocks. Research has shown that changes in economic policies can increase uncertainty in investors' expectations of future asset returns and risks, potentially having a significant impact on market stability. Machine learning models, such as LSTM networks, can be used to combine risk factors with text sentiment analysis to predict market sentiment, improving the accuracy of predictions.

Finally, considering the instability and volatility of financial markets, it may be beneficial to establish a deeper connection between the existing models and real-time financial market data. By leveraging deep learning techniques, the capability of sentiment analysis algorithms can be enhanced to better handle large amounts of unstructured data. Establishing a real-time market sentiment monitoring system would provide investors with immediate market sentiment analysis and risk alerts.

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Appendix

Appendix A: SPSS Operation Procedures

- Importing Data
- Data Standardization
- Click [Analyze]-[Descriptive Statistics]-[Descriptives]

The following prompt box appears, put the indicators on the left into the variables on the right, check [Save standardized values as variables].



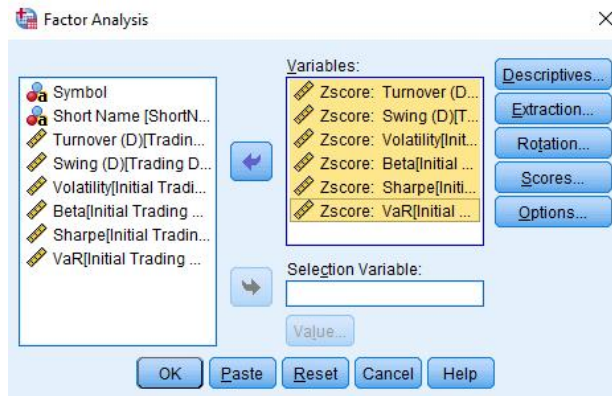
The following data will then appear.

ZTurnoverDTrading Date20240819Unit	ZSwingDTradingDa te20240819Unit	ZVolatilityInitialTra dingDate52Wbef.En dDateTradingEndD	ZBetaInitialTrading Date52Wbef.EndC ateTradingEndDate	ZSharpeInitialTrad ngDate52Wbef.En DateTradingEndDat	ZVaRInitialTrading Date52Wbef.EndC ateTradingEndDate
-.27550	-.24209	-.69923	.17373	.10191	-.70994
-.31449	-.35656	-.15511	.13618	.10191	-.13231
-.35197	-.19774	-.55689	.08722	.10190	-.55636
.14761	.14331	-.16846	.13540	.10190	-.14324
1.51258	1.72497	-.49134	.15103	.10194	-.50144
.83147	-.15964	-.15838	.02388	.10186	-.11583
.25484	1.06867	-.39153	-.06122	.10197	-.40765
.98668	.02580	-.11600	.14505	.10200	-.13635
-.33253	-.12566	-.64109	.10132	.10194	-.65586
-.01755	-.15065	-.35527	.11923	.10196	-.36578
.15591	.12680	-.23587	.07392	.10191	-.22150
-.11344	-.11572	-.22658	.24069	.10199	-.24653
-.10111	.12680	.09372	.09157	.10196	.09696
4.43278	3.50758	-.09474	.10330	.10198	-.10665

PCA(principal component analysis)

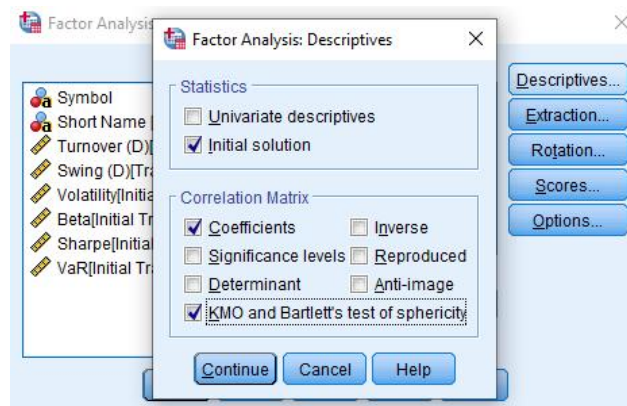
- Click [Analyze]-[Dimension Reduction]-[Eactor]

The following prompt box appears, put the standardized variable on the right



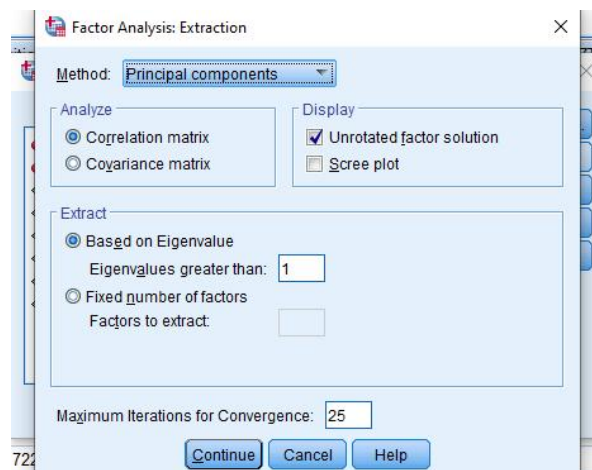
- Click [Descriptives]

The following prompt box appears, select the following.



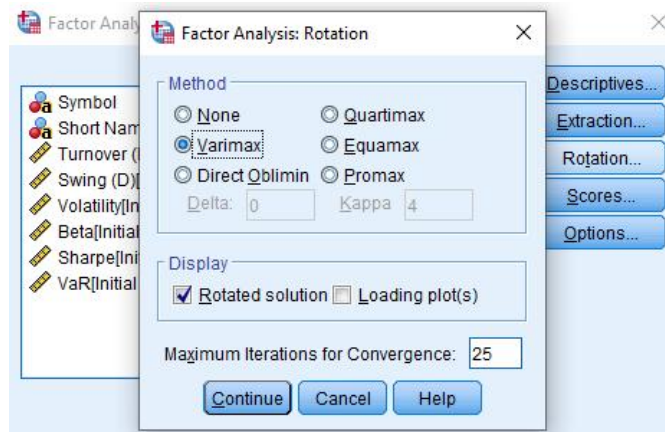
- Click [Extraction]

The following prompt box appears, select the following.



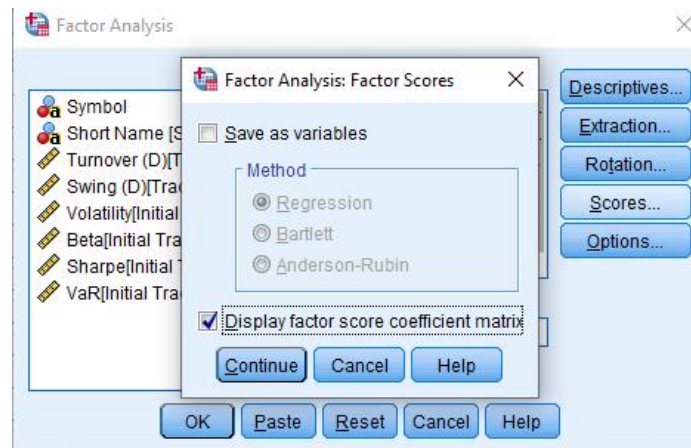
- Click [Rotation]

The following prompt box appears, select the following.



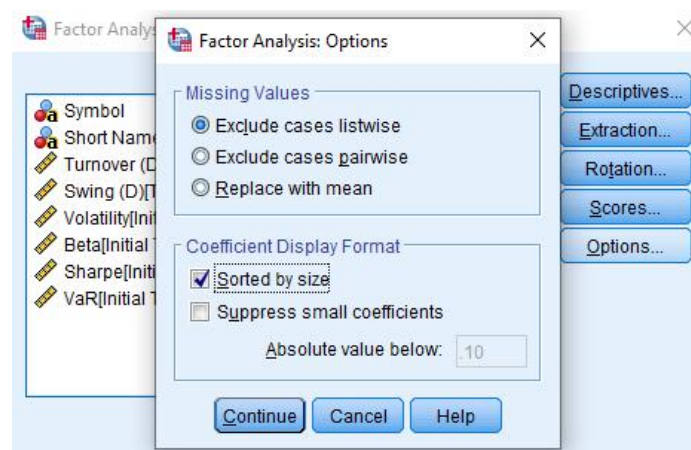
- Click [Scores]

The following prompt box appears, select the following.



- Click [Options]

The following prompt box appears, select the following.



- Click [Continue]

The results of principal component analysis were obtained (output_20240819).

Appendix B: Python Code for ARIMA

```
import pandas as pd
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import adfuller
import itertools
from statsmodels.tsa.arima.model import ARIMA
from pmdarima import auto_arima

### 1.Data Preprocessing and Visualization
# Read the CSV file
data = pd.read_csv('Variance.csv', parse_dates=['Date'], index_col='Date')
data = data.dropna(subset=['Variance'], axis=0)
data

# Draw a time series of stock closing prices and observe the trend, seasonality and volatility of the data.
plt.figure(figsize=(16,6))
plt.plot(data['Variance'], label='Variance')
plt.title('Variance Time Series')
plt.xlabel('Date')
plt.ylabel('Variance')
plt.legend()
plt.grid(True)
plt.show()

# Plot ACF
plt.figure(figsize=(12,6))
# ACF diagram
plot_acf(data['Variance'], lags=18, ax=plt.gca())
plt.title('Autocorrelation Function (ACF)')
plt.tight_layout()
plt.show()

# Observation: The data is not stable

### 2. Data Stabilization
# Augmented Dickey-Fuller (ADF)
```

```

# The ADF test is used to verify that the data is stable. If the P-value is greater than the significance level
(usually 0.05), the data is unstable and differential processing is required.
# Perform ADF test
adf_test = adfuller(data['Variance'])
# Output ADF test results
print('ADF Statistic:', adf_test[0])
print('p-value:', adf_test[1])
print('Critical Values:', adf_test[4])
# Determine whether the data is stable according to the P-value
if adf_test[1] > 0.05:
    print("The data is unstable and requires differential processing")
else:
    print("data stable")
# Difference: Stationary Data
# first difference
data_diff = data['Variance'].diff().dropna()
data_diff = pd.DataFrame(data_diff)
data_diff.columns = ['Differenced Variance']
# ADF test is performed again to verify whether the data after difference is stable
adf_test_diff = adfuller(data_diff)
print('ADF Statistic after differencing:', adf_test_diff[0])
print('p-value after differencing:', adf_test_diff[1])
if adf_test_diff[1] > 0.05:
    print("After the difference, the data is still unstable, and further difference is needed")
else:
    print("The data after difference is stable")
# Draw the time sequence diagram after the difference
plt.figure(figsize=(10,6))
plt.plot(data_diff, label='Differenced Variance')
plt.title('Differenced Variance Time Series')
plt.xlabel('Date')
plt.ylabel('Differenced Variance')
plt.legend()
plt.grid(True)
plt.show()
# Output difference result
data_diff

```

```

### 3. ACF and FACF
# Plot ACF and PACF diagrams
plt.figure(figsize=(12,6))
# Draw the ACF plot and use the data after difference
plt.subplot(1, 2, 1)
plot_acf(data_diff['Differenced Variance'], lags=18, ax=plt.gca())
plt.title('Autocorrelation Function (ACF)')
# Draw the PACF map and use the data after difference
plt.subplot(1, 2, 2)
plot_pacf(data_diff['Differenced Variance'], lags=18, ax=plt.gca())
plt.title('Partial Autocorrelation Function (PACF)')
plt.tight_layout()
plt.show()

### 4. Training Set and Test Set
# Split data into training set and test set (80% training set, 20% test set)
train_data, test_data = train_test_split(data, test_size=0.2, shuffle=False)
# View the size of the training and test sets
print(f"Training set size: {len(train_data)}")
print(f"Test set size: {len(test_data)}")
# Displays the first few lines of data for the training and test sets
print("The first few lines of the training set: ")
print(train_data.head())
print("Test the first few lines of the set: ")
print(test_data.head())

### 5. Parameter Selection and Model Training
# Set parameter range
p = d = q = range(0, 5) # Set the maximum value of p, d, and q to 5
pdq = list(itertools.product(p, d, q)) # Generates all possible (p, d, q) combinations
# Create an empty DataFrame to store the results
results = []
# Iterate over all possible parameter combinations
for param in pdq:
    try:
        # Fit ARIMA model
        model = ARIMA(train_data['Variance'], order=param)
        model_fit = model.fit()

```

```

# Get AIC and BIC values
aic = model_fit.aic
bic = model_fit.bic

# Calculate the AICc value
n = len(data)
k = len(model_fit.params) # Number of model parameters
aicc = aic + (2 * k * (k + 1)) / (n - k - 1) if n > k + 1 else np.nan

# Add the result to the list
results.append({
    'p': param[0],
    'd': param[1],
    'q': param[2],
    'AIC': aic,
    'BIC': bic,
    'AICc': aicc
})

print(f'ARIMA {param} - AIC: {aic}, BIC: {bic}, AICc: {aicc}')

except Exception as e:
    # Ignore parameter combinations that may not fit
    continue

# Convert the results to a DataFrame
results_df = pd.DataFrame(results)

# Sort by BIC value to find the optimal parameter combination
results_df = results_df.sort_values(by='BIC')

# Print the results and save to a CSV file
print(results_df)
results_df.to_csv('arima_model_results.csv', index=False)
results_df

### 6. Prediction and result visualization
# Use auto_arima to automatically select the best parameters
from pmdarima import auto_arima

# Automatically select the best ARIMA parameters based on AIC or BIC
auto_model = auto_arima(train_data['Variance'], seasonal=False, trace=True,
                        error_action='ignore', suppress_warnings=True, stepwise=True)

```

```

# Fit the best ARIMA model
model_fit = auto_model.fit(train_data['Variance'])

# Print the model summary
print(model_fit.summary())

# Make predictions for the test set and the next five steps
forecast_steps = len(test_data) + 5
forecast = model_fit.predict(n_periods=forecast_steps)

# Create forecast index
forecast_index = pd.date_range(start=test_data.index[0], periods=forecast_steps, freq='B')

# Plot the comparison between actual and forecasted values
plt.figure(figsize=(10, 6))
plt.plot(train_data.index, train_data['Variance'], label='Train Data')
plt.plot(test_data.index, test_data['Variance'], label='Test Data')
plt.plot(forecast_index, forecast, label='Forecast', color='red')
plt.title('Improved ARIMA Forecast')
plt.xlabel('Date')
plt.ylabel('Variance')
plt.legend()
plt.grid(True)
plt.show()

# Output model parameter
print(model_fit.params())

```